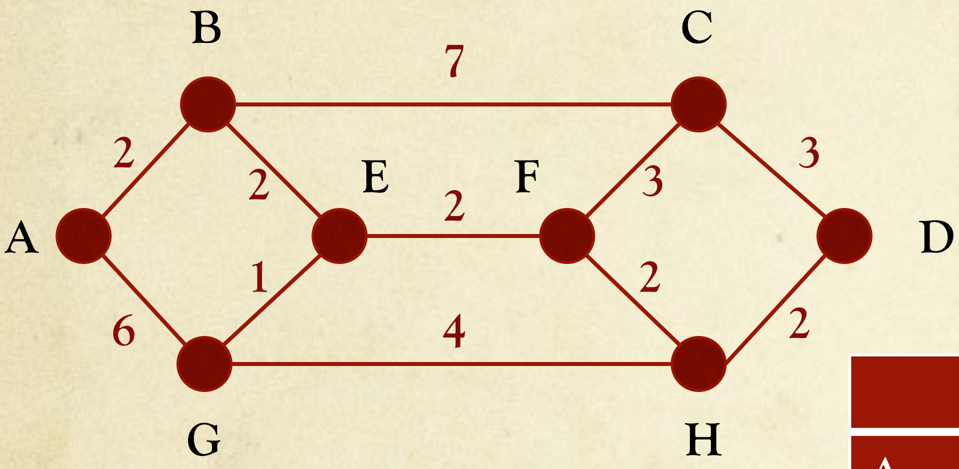
Shortest path routing

- Route packets along *shortest* path b/w src & dst
- Shortest path calculated & *stored* in routing table
- Shortest path could mean
  - Lowest number of hops
  - Shortest geographic distance
  - Lowest average delay (i.e., fastest path)
- Can assign *labels* using weighted "distance"

Djkstra's algorithm

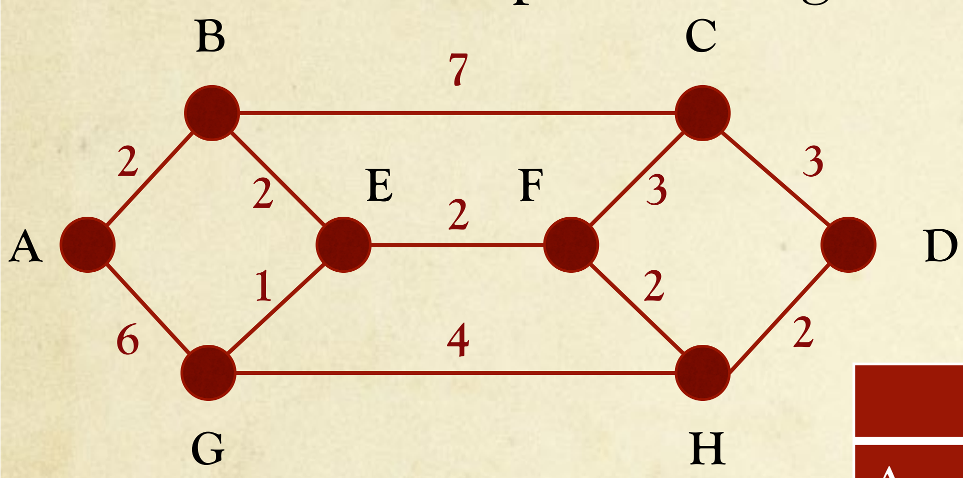
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A								

Djkstra's algorithm

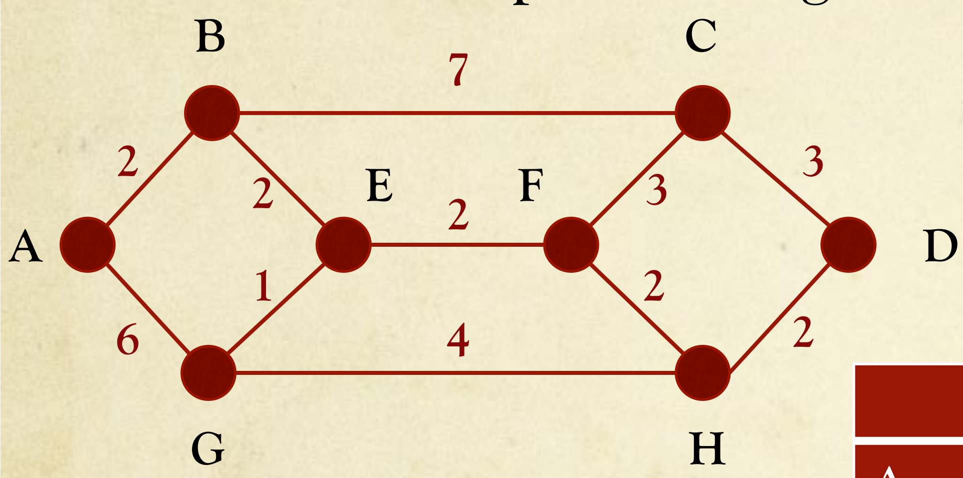
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B								

Djkstra's algorithm

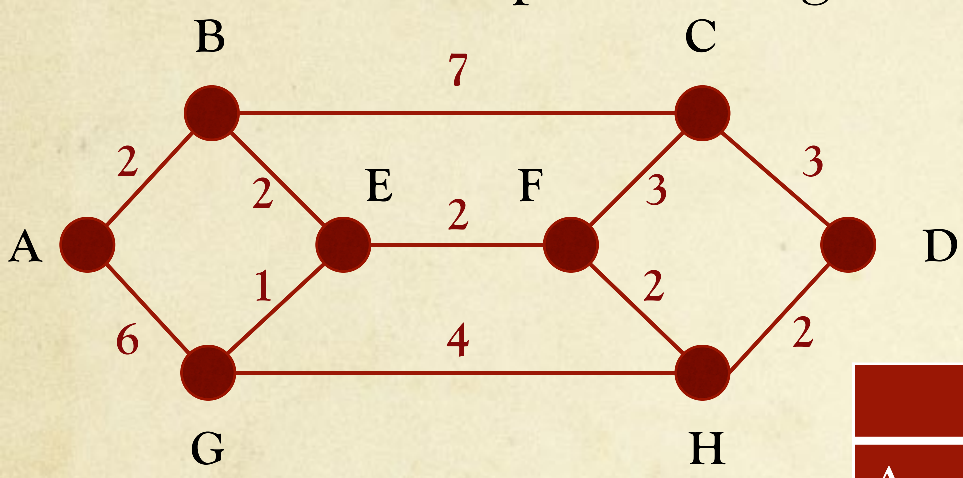
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B	0_A	2_A	9_B	∞_A	4_B	∞_A	6_A	∞_A
E								

Djkstra's algorithm

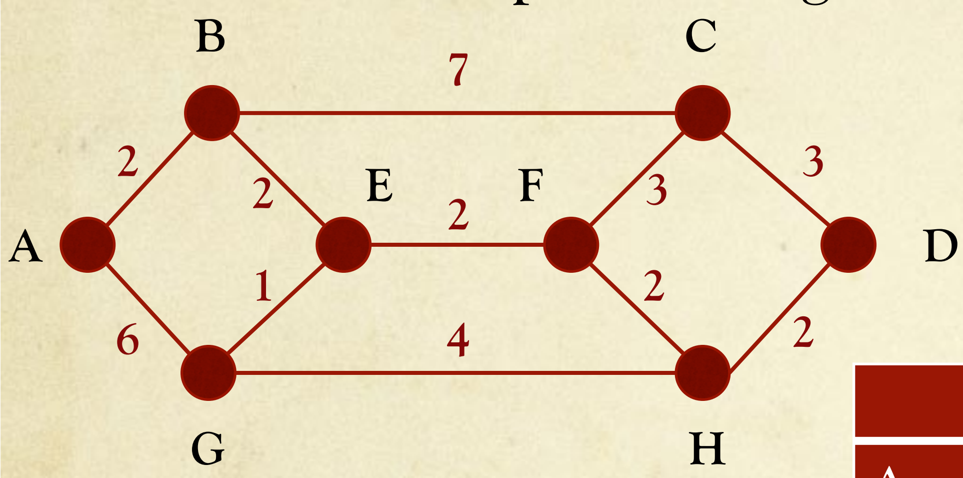
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B	0_A	2_A	9_B	∞_A	4_B	∞_A	6_A	∞_A
E	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	∞_A
G								

Djkstra's algorithm

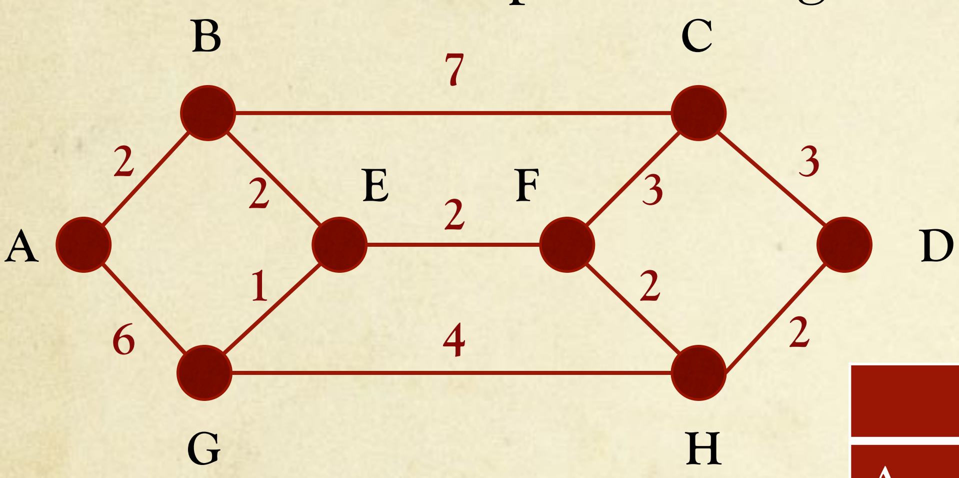
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B	0_A	2_A	9_B	∞_A	4_B	∞_A	6_A	∞_A
E	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	∞_A
G	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	9_G
F								

Djkstra's algorithm

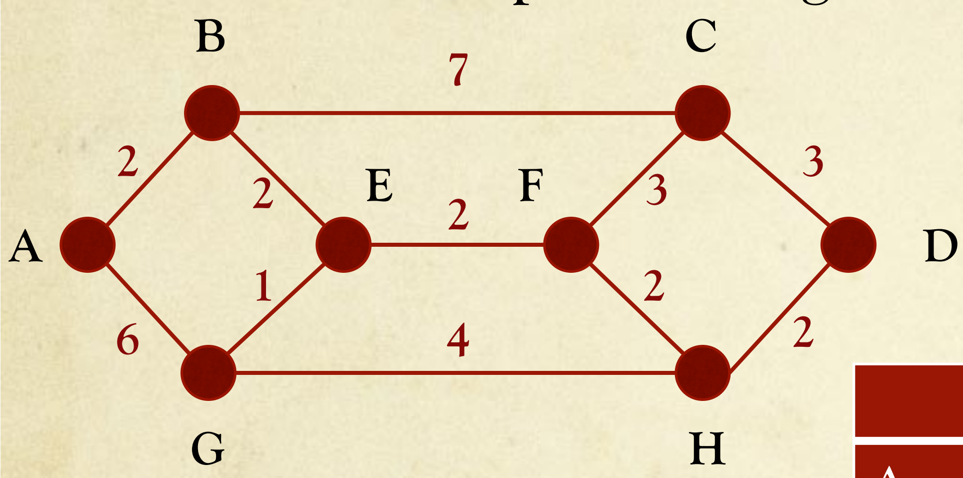
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B	0_A	2_A	9_B	∞_A	4_B	∞_A	6_A	∞_A
E	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	∞_A
G	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	9_G
F	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	8_F
H								

Djkstra's algorithm

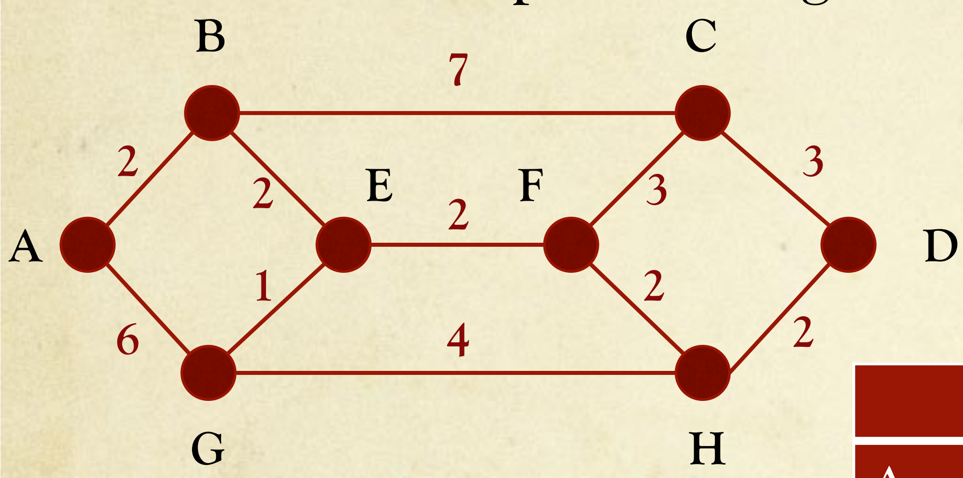
Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B	0_A	2_A	9_B	∞_A	4_B	∞_A	6_A	∞_A
E	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	∞_A
G	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	9_G
F	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	8_F
H	0_A	2_A	9_B	10_H	4_B	6_E	5_E	8_F
C								

Djkstra's algorithm

Finds *shortest* path b/w given *src* & *all* other nodes in n/w



	A	B	C	D	E	F	G	H
A	0_A	2_A	∞_A	∞_A	∞_A	∞_A	6_A	∞_A
B	0_A	2_A	9_B	∞_A	4_B	∞_A	6_A	∞_A
E	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	∞_A
G	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	9_G
F	0_A	2_A	9_B	∞_A	4_B	6_E	5_E	8_F
H	0_A	2_A	9_B	10_H	4_B	6_E	5_E	8_F
C	0_A	2_A	9_B	10_H	4_B	6_E	5_E	8_F
D								

Hierarchical routing

- As network size increases, routing tables get very *large*!
- Set up *hierarchy* among routers
  - Form regions
  - Each router knows details of its own region
  - Designated routers to get outside region
- Concept can be extended to *multiple* levels of hierarchy
  - Region, cluster, zone, ...

When are routing decisions made?

- *Connectionless* service [*widely used*]
  - Routing decisions made independently for every packet
  - Subsequent packets may or may not take same route
- *Connection-oriented* service
  - Routing decision made during connection establishment
  - Same route used for *all* packets on same connection